



CHAPTER

Tanks

Diluate, concentrate, electrolyte and drain tanks

IN THIS CHAPTER

15.1	Overview and function	102
15.2	Key specifications	103
15.3	Operation	104
15.4	Maintenance	105
15.5	Troubleshooting	106

15.1 Overview and function

Overhead/skid-mounted tanks buffer the diluate, concentrate and electrolyte streams (chapter 3) and collect drainage from both containers before it leaves the unit. In the pilot configuration these are tagged TSA (a sensor/buffer tank), TB and TE (electrolyte) — see chapter 13 for the level/conductivity instrumentation on each.



Figure 15.1. Suggested figure: photograph of the tank bank inside the treatment container, each tank labeled with its stream (diluate/ concentrate/electrolyte/drain).

15.2 Key specifications

No tank capacity/material specification exists yet for a finalized standard SalinBloc™ bill of materials, but two real, recent tank orders placed by HydroVolta give a concrete sense of the sizes used:

- Rotterdam Plastics order (Oct. 2025): one 800-litre vertical polyethylene tank (closed outlet, unvented lid) and one 450-litre polyethylene tank (open outlet, vented lid).
- Boralit/Galo Tanks order: two 1000-litre rectangular polyethylene tanks (RT1000RH ZW series), supplied with PVC coupling sets.

TODO: which physical tank (diluate/feed, concentrate, electrolyte, or drain) each of the above corresponds to was not confirmed in the order documents — do not assume a mapping without checking the as-built P&ID for the specific unit. The pilot configuration referenced elsewhere in this manual used a 1000 l IBC as an auxiliary/drain tank.

15.3 Operation

Tank levels are monitored continuously (TSA_Level, TB_Level, TE_Level) with high/low alarm limits — see appendix A.

15.4 Maintenance

TODO: no tank cleaning/inspection schedule was found.

15.5 Troubleshooting

High/low level alarms are confirmed for every tank: HL_Alarm_TD/LL_Alarm_TD (diluate), HL_Alarm_TC/LL_Alarm_TC (concentrate), HL_Alarm_TB/LL_Alarm_TB, HL_Alarm_TSA/LL_Alarm_TSA, HL_Alarm_TE/LL_Alarm_TE, plus a level alarm M050_Level_Alarm. On any level alarm, check the relevant level sensor per the bench-test procedure in section 13.4 before assuming a process fault.